Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3.

Multilingual support for template handling
Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation
Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types
Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers
TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

hyperref [T1]fontenc
pgfpages
latexsym,amsmath,xcolor,multicol,booktabs,calligra
graphicx,listings,stackengine
QUT
basicstyle=, keywordstyle=, emphstyle=, stringstyle=,
numbers=left, numberstyle=, rulesepcolor=,
frame=shadowbox,

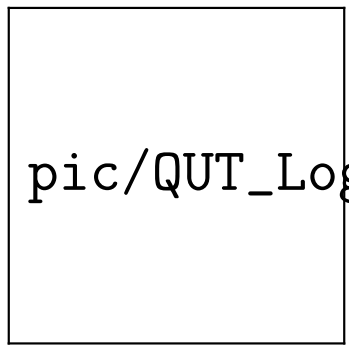
QUT Beamer Theme

Presentation

Presenter

School of Information Systems
Queensland University of Technology

19 Pebruari 2025



Introduction

Literature Review

Research Problem

Research Plan

References

Original Template

- Modify from this theme [?]

Original Template

- Modify from this theme [?]
- Overleaf https://www.overleaf.com/latex/templates/thu-beamer-theme/vwnqmzndvwyb

Original Template

- Modify from this theme [?]
- Overleaf `https://www.overleaf.com/latex/templates/thu-beamer-theme/vwnqmzndvwyb`
- GitHub Page `https://github.com/Trinkle23897/THU-Beamer-Theme`

Beamer Features

- More features come from
  `https://www.latexstudio.net/archives/4051.html`

Formatting Samples

Equation without numbers

$$J(\theta) = \mathbb{E}_{\pi_\theta}[G_t] = \sum_{s \in \mathcal{S}} d^\pi(s) V^\pi(s) = \sum_{s \in \mathcal{S}} d^\pi(s) \sum_{a \in \mathcal{A}} \pi_\theta(a|s) Q^\pi(s, a)$$

Multiple equations[1]

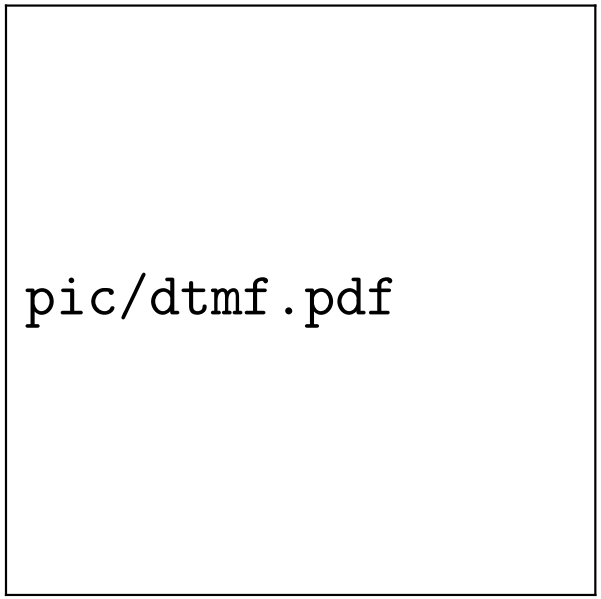
$$Q_{\text{target}} = r + \gamma Q^\pi(s', \pi_\theta(s') + \epsilon) \tag{1}$$
$$\epsilon \sim \text{clip}(\mathcal{N}(0, \sigma), -c, c)$$

[1] If containing text in equationsïijŇuse \mathrm{} or \text{}

Equation with numbers

$$
\begin{aligned}
A = \lim_{n\to\infty} \Delta x \Big(& a^2 + \left(a^2 + 2a\Delta x + (\Delta x)^2\right) \\
& + \left(a^2 + 2\cdot 2a\Delta x + 2^2(\Delta x)^2\right) \\
& + \left(a^2 + 2\cdot 3a\Delta x + 3^2(\Delta x)^2\right) \\
& + \dots \\
& + \left(a^2 + 2\cdot (n-1)a\Delta x + (n-1)^2(\Delta x)^2\right)\Big) \\
& = \frac{1}{3}\left(b^3 - a^3\right) \quad (2)
\end{aligned}
$$

Figure and Column



LaTeX Commands

Commands

\chapter	\section	\subsection	\paragraph
Chapter	Section	Subsection	Paragraph
\centering	\emph	\verb	\url
Centre Align	Emphasis	Verbatim	Hyperlink
\footnote	\item	\caption	\includegraphics
Foodnote	Item	Caption	FigP&Pic
\label	\cite	\ref	
Label	Citing	Referring	

Environment Command

table	figure	equation
Table	Figure	Equation
itemize	enumerate	description
Bullets	Numbering	Description

LaTeX Environment Command Samples

[language=TeX]

- A
- B
- C
  - C-1

- A
- B
- C
  - C-1

LaTeX Environment Command Samples

[language=TeX]

- A
- B
- C
  - C-1

[language=TeX]

1. Class 1
2. Class 2
3. Class 2

   n+e Student 1

- A
- B
- C
  - C-1

1. Class 1
2. Class 2
3. Class 3

   n+e Student 1

LaTeX Equations

[language=TeX] $V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi r^3$

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3 \tag{3}$$

$V = \frac{4}{3}\pi r^3$

$V = \frac{4}{3}\pi r^3$

$$V = \frac{4}{3}\pi r^3$$

$$V = \frac{4}{3}\pi r^3 \tag{4}$$

- Check more Here

[language=TeX]

Tabel: Definition

Word	Definition
1	4.0
2	3.7

Check definition of Equation (4) in Table 2ãĂĆ

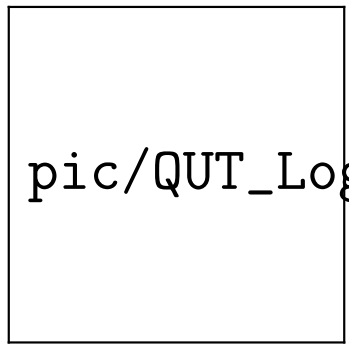
Tabel: Definition

Eq.	Def.
1	4.0
2	3.7

Please check the definition of Equation (4) in Table 2

Plotting

- Vector: eps, ps, pdf
  - METAPOST, pstricks, pgf ...
  - Xfig, Dia, Visio, Inkscape ...
  - Export Matlab / Excel as pdf
- Bitmap: png, jpg, tiff ...
  - Avoiding using bitmaps



Gambar: This is a Bitmaps

- Year 1
- Year 2
- Year 3
- ...

Thanks!